

Refreshing Evaporative Coolers

California's energy woes have focused new attention on energy-saving technologies long considered underutilized. High on the list is evaporative cooling, which lowers the temperature of air by passing it through water-soaked pads.

A number of Western utilities are promoting this long-out-of-favor technology through evaporative cooler rebates that can cover 10%–15% of the typical cost of room-sized, generally window-mounted equipment. Southern California Edison offers a \$100 rebate on evaporative coolers that are permanently installed in conjunction with an existing central air conditioning unit or heat pump. And Pacific Gas and Electric (PG&E) has announced a \$300 rebate on a whole-house model that meets specific efficiency standards (see "The PG&E Rebate Program," p. 21).

Evaporative Cooling

Evaporative cooling works because changing water from liquid to gas—evaporation—requires energy (heat). In the simplest, "single stage," form of evaporative cooling, dry air passing through a cooler's water-soaked pads picks up water through evaporation; the energy to accomplish evaporation comes directly from the heat in the incoming air (see Figure 1). A fan or blower forces the airstream—now both cooler and damper—into an overheated space, displacing an equal amount of warmer air, which must exit through an open window or vent.

In two-stage evaporative cooling, outdoor air first passes inside a heat exchanger, cooled on the outside by evaporation. This precooled air then enters a second chamber that works in the same way as the single-stage

evaporative coolers. Manufacturers of evaporative coolers produce and sell attachments to one-stage coolers to create two-stage coolers. (See "Two-Stage Evaporative Cooling," *HE* May/June '97, p. 31.)

Two-stage evaporative coolers

holds using evaporative coolers did not keep up with general population growth; penetration fell to 3% of total households by 1993. During that same time period, the penetration of central air conditioning systems almost doubled—from

27% of all households in 1980 to 44% in 1993—and the number of households without any type of air conditioning fell from 43% to 32%.

The move away from evaporative coolers could be due to a number of factors, such as the relatively high indoor humidity levels created in evaporative cooling, or the maintenance required. Perhaps the larger manufacturers are more comfortable making an air conditioning product that can be used across the country, rather than making evaporative coolers only for regional markets.

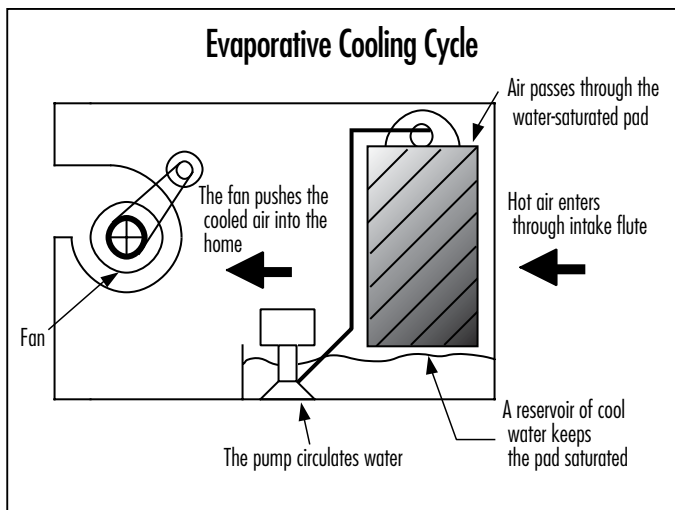


Figure 1. Have you ever fanned yourself or stood in a breeze while sweating on a hot day? If so, you've experienced the basic cause (evaporation of water) and effect (cooling) of evaporative cooling.

lower indoor air temperatures a few more degrees, create drier indoor air, and can handle outdoor air humidity a little bit better than one-stage coolers. For example, in Phoenix, two-stage evaporative coolers meet ASHRAE comfort standards for school buildings at 3,500 ft elevation and higher. Single-stage coolers meet the standards at 5,000 ft elevation and higher, where the humidity is relatively lower.

High outdoor humidity levels make evaporative cooling inefficient and unsuitable for most of the United States east of the Mississippi, as well as for the northern and central West Coast (see Figure 2). However, in drier regions the technology has eased many Western families through hot weather for decades in the form of the window-mounted "swamp coolers." By 1980, the first year for which the Energy Information Agency reports evaporative cooler use, 4% of U.S. residences used them. But the number of house-

Pluses

There's no doubt about the potential for evaporative cooling to save money and reduce electric energy use—30%–90% savings when compared with conventional, refrigerant-based air conditioning, according to the Partnership for Advanced Technology in Housing (PATH). In Southern California, Southern California Edison estimates that window units alone can reduce air conditioning bills by \$100–\$500 per season. Equipment costs are also considerably lower than costs for traditional air conditioners, as are replacement parts. (See "Installing and Maintaining Evaporative Coolers," *HE* May/June '96, p. 23.)

Evaporative coolers are also environmentally friendly. They don't use CFCs or other refrigerants that are found in standard air conditioning systems. And advances in technology in the last 50 years have made evaporative coolers less aggra-

vating to operate. Mid twentieth-century cooler technology left a great deal to be desired:

- Coolers required high levels of maintenance.
- The water-holding aspen pads (made from the thin wood strips commonly known as excelsior) required frequent changing.
- Malfunctioning evaporative coolers

leaked, with window-mounted units sometimes dripping water on the heads of passersby.

- During extended periods when the coolers weren't in use, they frequently grew mold when water pooled in the drip pan or reservoir, creating unpleasant smells and aggravating allergies.
- Minerals from hard water and dirt from the air accumulated inside the units, reducing efficiency.
- One-speed fans could easily make

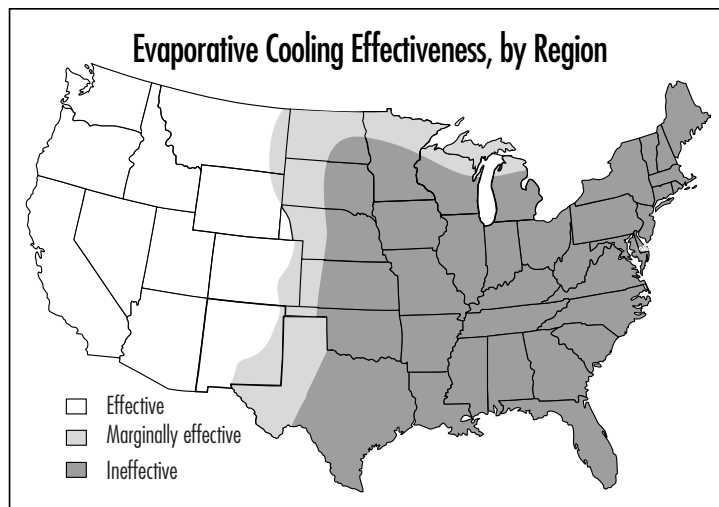


Figure 2. In areas of the United States where humidity tends to run high in the summertime, evaporative cooling is not the best way to stay cool; in the West, it's a very good choice.

rooms too cold, especially if the cooler had no thermostat.

- The constant contact of water and metal caused rapid rust-out.
- New units avoid these problems in the following ways:

- Rigid pads of manufactured materials last far longer than the old aspen pads. Phoenix Manufacturing, for instance, says its rigid pads last up to ten times longer than aspen models.
- Sump pumps have replaced bleed systems, removing water more effectively and actively purging systems of minerals and dirt at specified intervals.
- Two-speed fans increase both comfort and efficiency.
- Plastic housings, powder paints, and isolation of metal parts from water have dramatically reduced corrosion problems.
- The emergence of new evaporative coolers with thermostats—some that select speeds and one-stage or two-stage cooling modes—improve comfort and efficiency and therefore improve public acceptance of evaporative coolers.

Minuses

While evaporative cooling technology has advanced since the 1950s, there are still problems to be solved.

Here are some of those problems:

You need to start with dry air that is not too hot. Adobe Air claims that its evaporative coolers can deliver 78°F indoor air temperature with an outdoor air temperature of 88°F and 54% relative humidity (or with outside air at 106°F and 16% relative humidity). That's not a problem during most of the year in the Southwest and West, but it definitely is a problem in the East, where heat and humidity go hand in hand. Thus, while refrigeration-based air conditioner

manufacturers enjoy a nationwide market, evaporative cooler manufacturers have fewer potential customers and must spread development and manufacturing costs over fewer units.

Evaporative cooling adds considerable humidity to the air. Homes with one-stage coolers typically experience humidity levels of about 80%, according to PATH, making many people uncomfortable.

The technology uses a considerable amount of water. Single-stage evaporative coolers use 3 to 11 gallons per day or more, which must be figured into operating costs.

Roof installation can be a logistical headache. While evaporative coolers are not as heavy as air conditioning equipment, hauling units up several stories may not be easy. You must also cut a hole in the roof. Roof installation—the most popular type—complicates maintenance and also decreases efficiency by about 1% on sunny rooftops. Some users also report increased roof deterioration as a result of increased foot traffic and leaking coolers.

Ducting is required for whole-house models. This adds to total costs. Many homeowners attempt to avoid that cost by connecting their system to preexisting ductwork. Ideally, evaporative coolers would be provided with larger ducts than are

The PG&E Rebate Program

PG&E is one of the utilities offering rebates (\$300, through the end of 2001) on whole-house evaporative coolers. To qualify for the rebate, homeowners must purchase equipment that meets the following specifications:

- Air flow must be 2,500 CFM or greater at 0.1-inch static pressure.
- The evaporative media must be rigid and must achieve a saturation efficiency of 85% or better.
- The unit must have at least two fan speeds.
- The controls must permit the unit to operate with the media wet or dry.
- The units must be UL listed.

Rebates are not available in California Energy Commission climate zones 1 and 3 (northern and central coast).

typically installed for traditional home cooling and heating systems. Ducts that are too small result in lower system efficiencies.

Care must be taken to locate the cooler's air intake away from vents or flues. Vents for furnaces and openings for appliances like dryers can lead to nonoptimal installations, building code violations, and health and safety problems. It may be costly and difficult to relocate these vents, but it must be done.

Homeowners may require both evaporative cooling and traditional air conditioning to maintain year-round comfort. Areas like Arizona and Nevada, where evaporative cooling works well over most of the year, can experience short periods of wet or intensely hot weather with which evaporative cooling can't cope. In areas like Phoenix, evaporative cooling systems generally save enough in electricity costs to justify the cost of installing and maintaining two systems. That may not be the case in

areas like Las Vegas, however, where electricity rates are traditionally low, reducing the overall cost of operating a traditional air conditioning system and making it more difficult to cost justify the addition of evaporative cooling.

Look Twice

In areas where hot weather brings dry air, consumers can almost always reduce their electricity use by using a low-cost window evaporative cooler. "Personal" models are also available; they are very small and are typically located on a nearby desk or table.

Whole-house installations may also be worthwhile, but homeowners will want to make sure that the resulting relatively high levels of interior humidity will prove comfortable to family members and will not exacerbate the growth of molds that can aggravate allergies or damage furniture, books, or clothing. You'll also want to calculate the cost-benefit ratio

carefully. Equipment costs are considerably lower for whole-house evaporative coolers than they are for central air conditioners, but equipment costs aren't the entire story. You may get higher cost-benefit ratios for efficiency measures like insulating or weatherstripping, especially for energy-inefficient homes in areas where hot-weather humidity varies considerably. 

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For more information:

Information about the PG&E and Southern California Edison rebate programs can be found at www.pge.com/123 and www.sce.com, respectively.